



Maryland Comprehensive
Assessment Program

Grade 3
Mathematics
Practice Test



Maryland
STATE DEPARTMENT OF EDUCATION

If you do not know the answer to a question, you may go on to the next question. If you finish early, you may review your answers and any questions you did not answer in this Section **ONLY**. Do not go past the stop sign.

Directions for Completing the Answer Grids

1. Work the problem and find an answer.
2. Write your answer in the boxes at the top of the grid.
3. Print only one number or symbol in each box. Do not leave a blank box in the middle of an answer.
4. Under each box, fill in the circle that matches the number or symbol you wrote above. Make a solid mark that completely fills the circle.
5. Do not fill in a circle under an unused box.
6. See below for examples on how to correctly complete an answer grid.

EXAMPLES

To answer 632 in a question, fill in the answer grid as shown below.

6	3	2			
☐	☐	☐	☐	☐	☐
0	0	0	0	0	0
1	1	1	1	1	1
2	2	●	2	2	2
3	●	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
●	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

A brick path has 10 rows of 4 bricks. How many bricks are in the path?

Enter your answer in the box.

4	0				
☐	☐	☐	☐	☐	☐
0	●	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
●	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

- The figure shows two 2x4 grids. The top grid has shaded squares at positions (1,1), (1,3), (2,2), and (2,4). The bottom grid has shaded squares at positions (1,1), (1,2), (2,1), and (2,4).

- GO ON ►**

- 3 The floor of a school hallway is in the shape of a rectangle.
The length of the floor is 8 feet and the width of the floor is 6 feet.

What is the area of this hallway?

- Ⓐ 14 square feet
- Ⓑ 28 square feet
- Ⓒ 48 square feet
- Ⓓ 56 square feet

- 4 Which list shows how $\frac{2}{4}$ compares to $\frac{2}{6}$ and $\frac{2}{8}$?

- Ⓐ $\frac{2}{4} < \frac{2}{6}$ and $\frac{2}{8} < \frac{2}{4}$
- Ⓑ $\frac{2}{4} > \frac{2}{6}$ and $\frac{2}{8} < \frac{2}{4}$
- Ⓒ $\frac{2}{4} < \frac{2}{6}$ and $\frac{2}{8} > \frac{2}{4}$
- Ⓓ $\frac{2}{4} > \frac{2}{6}$ and $\frac{2}{8} > \frac{2}{4}$

5 There are 63 cars parked in a parking lot.

- There are 7 rows in the parking lot.
- Each row has the same number of parked cars.

How many cars are parked in each row?

Enter your answer in the space provided.

●	●	●	●	●	●
0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

6 Two students drew rectangles.

The first student drew a rectangle with a length of 6 inches and a width of 4 inches.

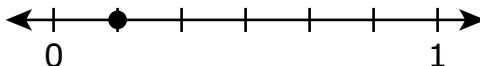
The second student drew a rectangle with the same perimeter as the rectangle the first student drew.

The width of the rectangle the second student drew is 7 inches.

What was the length, in inches, of the rectangle the second student drew?

- Ⓐ 3
- Ⓑ 6
- Ⓒ 13
- Ⓓ 17

- 7 What fraction is represented by the point on the number line?



- (A) $\frac{6}{6}$
- (B) $\frac{5}{6}$
- (C) $\frac{2}{6}$
- (D) $\frac{1}{6}$
- 8 Which statement is true?
- (A) When 6 is added to an odd number, the sum will be an even number.
- (B) When 7 is added to an even number, the sum will be an odd number.
- (C) When 5 is multiplied by an odd number, the product will be an even number.
- (D) When 4 is multiplied by an even number, the product will be an odd number.
- 9 Which number rounds to 300 when rounded to the nearest hundred?
- (A) 216
- (B) 241
- (C) 349
- (D) 384

-
- Four identical glass jars are shown in a row. Each jar has a white screw-on lid and contains 10 small, round, gray beads. The beads are scattered at the bottom of each jar.

Ⓐ 4×8

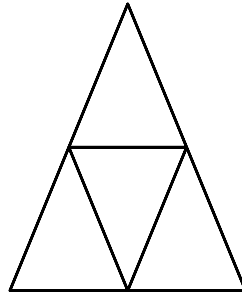
Ⓑ $4 + 8$

Ⓒ $4 + 4 + 4 + 4$

Ⓓ $8 + 8 + 8 + 8 + 8 + 8 + 8 + 8$

- Ⓐ 2
- Ⓑ 3
- Ⓒ 9
- Ⓓ 18

- 12** The triangle shown represents one whole and has been partitioned into smaller triangles of equal size.



What fraction of the larger triangle does each smaller triangle represent?

- Ⓐ $\frac{3}{4}$
- Ⓑ $\frac{2}{3}$
- Ⓒ $\frac{1}{3}$
- Ⓓ $\frac{1}{4}$



Section 2

Today, you will take Section 2 of the Grade 3 Mathematics Practice Test. You will be allowed to use a calculator.

Read each question. Then, follow the directions to answer each question. Mark your answers by completely filling in the circles in your test book. Do not make any pencil marks outside of the circles. If you need to change an answer, be sure to erase your first answer completely. If a question asks you to show or explain your work, you must do so to receive full credit. Only responses written within the space provided will be scored.

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3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
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7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

A brick path has 10 rows of 4 bricks. How many bricks are in the path?

Enter your answer in the box.

4	0				
☐	☐	☐	☐	☐	☐
0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9



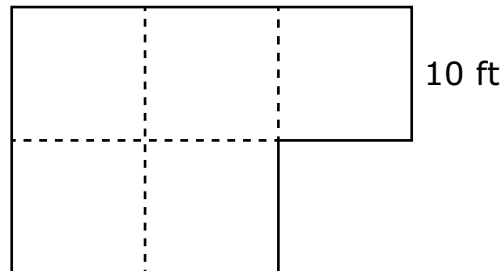
1 A teacher is working with 4 students on an art project. The teacher opened a package of paper and gave each student 9 sheets of paper.

Which question can be answered using **only** this information?

- Ⓐ What is the total number of sheets of paper left in the package?
- Ⓑ What is the total number of sheets of paper the teacher gave to the students?
- Ⓒ What is the total number of students in the teacher's class who did not receive sheets of paper?
- Ⓓ What is the total number of sheets of paper that were in the package before the teacher opened it?



- 2 A person will cover a floor with carpet. The shape of the floor is shown. Each of the squares has a side length of 10 feet. The side of one square is labeled.



The person said 25 square feet of carpet are needed to cover the floor.

The person is incorrect.

Which statement explains how the person could correctly find the area of the floor?

- Ⓐ There are 5 squares each with a side length of 10 feet, so the person could multiply 5 by 10.
- Ⓑ There are 5 squares each with a perimeter of 40 feet, so the person could multiply 5 by 40.
- Ⓒ There are 5 squares each with an area of 5 square feet, so the person could multiply 5 by 5.
- Ⓓ There are 5 squares each with an area of 100 square feet, so the person could multiply 5 by 100.



Section 2

3

The student organized the pennies into the array shown.



The student wants to put all the pennies in stacks with the same number of pennies in each stack.

Explain three different ways the student could put the pennies in stacks with the same number of pennies in each stack.

Explain how you used the array of pennies to help you find the different ways to stack the pennies.

Enter your answer and your work or explanation in the space provided.



- 4 A student had 10 pieces of gum. The student gave 2 pieces of gum to some friends.

Which expression can be used to find the number of pieces of gum the student has left?

- Ⓐ $10 + 2$
- Ⓑ 10×2
- Ⓒ $10 - 2$
- Ⓓ $10 \div 2$



The student used the steps shown to find how many pieces of candy are left.

Step two: $5 \times 9 = 45$

Check the student's work.

Enter your answer and your work or explanation in the space provided.



6 A student made 20 necklaces.

- The student put 5 beads on each necklace.
- There were 10 beads in each package.

Which statement explains the student's correct thinking to find the total number of beads the student used to make all the necklaces?

- Ⓐ The student thinks that there are 20 necklaces and there are 10 beads in each package, and $20 \div 10 = 2$.
- Ⓑ The student thinks that there are 20 necklaces and there are 10 beads in each package, and $20 \times 10 = 200$.
- Ⓒ The student thinks that there are 20 necklaces and there are 5 beads on each necklace, and $20 \div 5 = 4$.
- Ⓓ The student thinks that there are 20 necklaces and there are 5 beads on each necklace, and $20 \times 5 = 100$.



Section 3

(Non-Calculator)

Directions:

Today, you will take Section 3 of the Grade 3 Mathematics Practice Test. You will not be allowed to use a calculator.

Read each question. Then, follow the directions to answer each question. Mark your answers by completely filling in the circles in your test book. Do not make any pencil marks outside of the circles. If you need to change an answer, be sure to erase your first answer completely. If a question asks you to show or explain your work, you must do so to receive full credit. Only responses written within the space provided will be scored.

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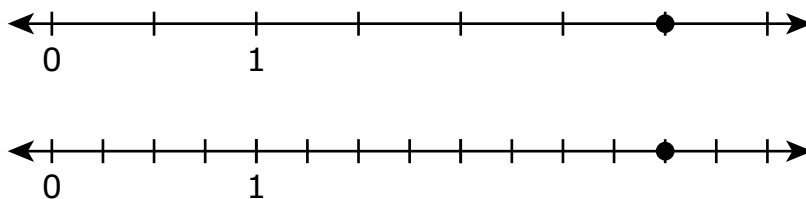
To answer 632 in a question, fill in the answer grid as shown below.

6	3	2			
●	●	●	●	●	●
0	0	0	0	0	0
1	1	1	1	1	1
2	2	●	2	2	2
3	●	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
●	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

Enter your answer in the box.

4	0				
●	●	●	●	●	●
0	●	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
●	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

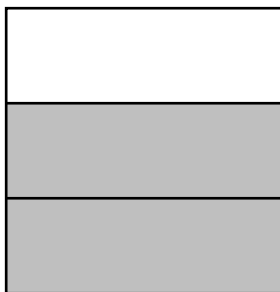
- 1 The points on the number lines shown represent equivalent fractions.



Which equation is represented by the points on the number lines?

- Ⓐ $\frac{6}{1} = \frac{12}{2}$
 - Ⓑ $\frac{6}{2} = \frac{12}{4}$
 - Ⓒ $\frac{6}{3} = \frac{12}{6}$
 - Ⓓ $\frac{6}{6} = \frac{12}{12}$
- 2 Which equation is correct?
- Ⓐ $30 \div 6 = 5$
 - Ⓑ $27 \div 3 = 8$
 - Ⓒ $36 \div 6 = 7$
 - Ⓓ $45 \div 5 = 8$

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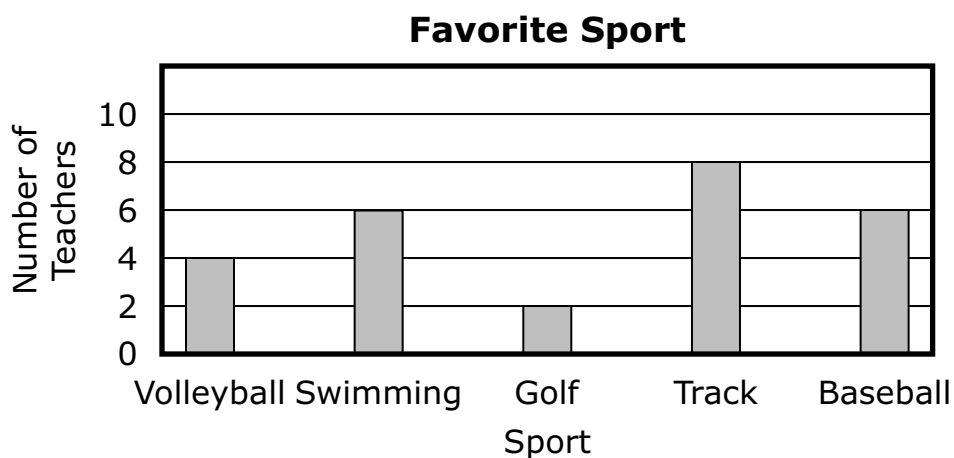


- 5 A teacher arranged 42 bottle caps into 7 rows with the same number of bottle caps in each row.

Which expression can be used to find the number of bottle caps the teacher put in each row?

- (A) $42 \div 7$
- (B) $42 \div 6$
- (C) 42×7
- (D) 42×6

- 6 Teachers at a school voted for their favorite sport. The bar graph shows the number of teachers who voted for each sport.



How many more teachers voted for track than voted for volleyball and golf combined?

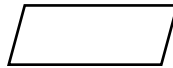
- (A) 1
- (B) 2
- (C) 4
- (D) 6

- 7** What is the value of $353 + 294$?

Enter your answer in the space provided.

●	●	●	●	●	●
0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

- 8** A piece of a whole model is shown. The piece represents $\frac{2}{6}$ of the whole model.



How many pieces like the one shown are in the whole model?

- Ⓐ 2
- Ⓑ 3
- Ⓒ 4
- Ⓓ 6

- 9 An expression is shown.

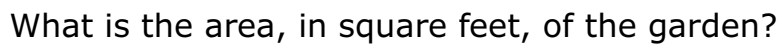
$$3 \times 10$$

Which **two** expressions have the same value as the expression shown?

Select the **two** correct answers.

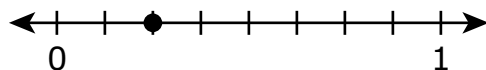
- Ⓐ $3 \times (2 + 5)$
- Ⓑ $3 \times (5 + 5)$
- Ⓒ $(3 \times 2) + (3 \times 5)$
- Ⓓ $(3 \times 5) + (3 \times 5)$
- Ⓔ $(3 \times 5) \times (3 \times 5)$

- The figure is composed of two rectangles that do not overlap.



- Ⓐ 38
- Ⓑ 61
- Ⓒ 72
- Ⓓ 90

- 11 The point on the number line represents a fraction.



Which fraction is represented by the point on the number line?

- Ⓐ $\frac{2}{8}$
- Ⓑ $\frac{2}{6}$
- Ⓒ $\frac{3}{6}$
- Ⓓ $\frac{6}{8}$



Section 4

(Calculator)

Directions:

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6	3	2			
●	●	●	●	●	●
0	0	0	0	0	0
1	1	1	1	1	1
2	2	●	2	2	2
3	●	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
●	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

Enter your answer in the box.

4	0				
●	●	●	●	●	●
0	●	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
●	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

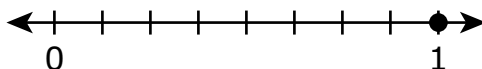


- 1** A gym teacher took some bags of soccer balls to the soccer field.
The teacher put 6 soccer balls in each bag.

What other information is needed to find how many bags of soccer balls the teacher took to the field?

- Ⓐ the total number of students in the gym class
- Ⓑ the total number of players on the school's soccer team
- Ⓒ the total number of soccer balls the teacher took to the field
- Ⓓ the total number of soccer balls each group of students will get

- 2** A point is shown on the number line.



Which **two** equations are true based on the values represented by the point on the number line?

Select the **two** correct answers.

- Ⓐ $\frac{1}{1} = 1$
- Ⓑ $\frac{1}{1} = 8$
- Ⓒ $\frac{1}{8} = 1$
- Ⓓ $\frac{1}{8} = 8$
- Ⓔ $\frac{8}{8} = 1$
- Ⓕ $\frac{8}{8} = 8$

- Enter your answer and your work or explanation in the space provided.

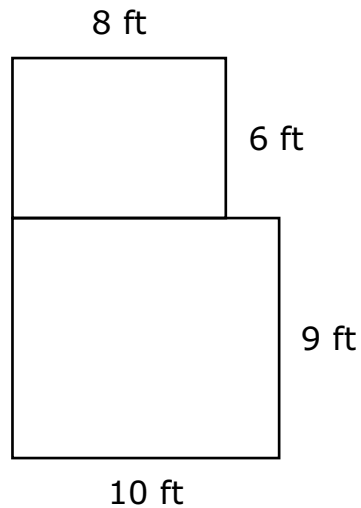


- 4** A farmer has a field with 4 rows of apple trees. There are 9 trees in each row.

Which expression could be used to find the total number of apple trees the farmer has in the field?

- Ⓐ $9 + 4$
- Ⓑ 9×4
- Ⓒ $9 - 4$
- Ⓓ $9 \div 4$

- 5** A pet store owner built two dog houses next to each other.
- The picture shows the rectangular floors of the two dog houses.



The pet store owner will cover the floors of the dog houses with concrete.

Determine the total area, in square feet, of the floors of the two dog houses. Be sure to show your work or explain how you found your answer.

Enter your answer and your work or explanation in the space provided.



- 6 A student wrote the expression $7 + 7 + 7 + 7 + 7$ to find the value of 7×5 .

The student said that when writing a different expression to represent 7×5 , only the numbers 7 and 5 can be used.

Which **two** expressions could be used to show that the student's response is incorrect?

Select the **two** correct answers.

- Ⓐ 5×7
- Ⓑ $(4 + 5) \times (3 + 5)$
- Ⓒ $(4 \times 5) + (3 \times 5)$
- Ⓓ $(7 + 3) \times (7 + 2)$
- Ⓔ $(7 \times 3) + (7 \times 2)$



The following pages include the answer keys for all machine-scored items. A sample student response for the top score is included for all hand-scored constructed response items.

- Some answer keys include one possible sample student response. Other valid methods for solving the problem can earn full credit unless a specific method is required by the item.
- In items where the scores are awarded for full and partial credit, the definition of partial credit will be confirmed during range-finding (reviewing sets of real student work).
- If students make a computation error, they can still earn points for reasoning or modeling.

Section 1

Item Number	Answer Key	Evidence Statement Key/ Content Scope
1.	A	3.NF.A.3b
2.	C, E	3.OA.C.7-1
3.	C	3.MD.C.7b
4.	B	3.NF.A.3d
5.	9	3.OA.A.3-2
6.	A	3.MD.D.8
7.	D	3.NF.A.2a
8.	B	3.OA.D.9
9.	C	3.NBT.A.1
10.	A	3.OA.A.1
11.	D	3.MD.A.2
12.	D	3.G.A.2

Section 2

Item Number	Answer Key	Evidence Statement Key/ Content Scope
1.	B	3.M.1 3.OA.A.3-1
2.	D	3.R.2 3.MD.C.7d
3.	<u>Sample Top Score Response</u> There are 4 rows of pennies in the array, so the student could make 4 stacks of pennies. Since there are 5 pennies in each row, there would be 5 pennies in each stack. There are 5 columns of pennies in the array, so the student could make 5 stacks of pennies. Since there are 4 pennies in each column, there would be 4 pennies in each stack. If I divide the array in half between the second and third rows, there would be 10 pennies in the top two rows and 10 pennies in the bottom two, so the student could make 2 stacks of pennies with 10 pennies in each stack. Refer to the Holistic Rubric for 3-Point Reasoning Constructed Response Items for score point information.	3.R.1 3.OA.A.2
4.	C	3.M.3 3.NBT.A.2

Item Number	Answer Key	Evidence Statement Key/ Content Scope
5.	<u>Sample Top Score Response</u> Step one shows the total number of pieces of candy the student has. Since there are 3 packages with 20 pieces of candy in each package, the numbers should be multiplied to find the total number of pieces of candy. Step two shows the total number of pieces of candy the student put in the treat bags. Since there are 5 treat bags and the student put 9 pieces of candy in each treat bag, the numbers should be multiplied to find the total number of pieces of candy the student put in the treat bags. Step three shows how many pieces of candy the student has left. Since the student started with 60 pieces of candy and put 45 pieces of candy in the treat bags, these numbers should be subtracted to find the total number of pieces of candy the student has left. Refer to the Holistic Rubric for 3-Point Modeling Constructed Response Items for score point information.	3.M.5 3.OA.D.8 3.NBT.A.2 3.NBT.A.3
6.	D	3.R.4 3.OA.A.3-1

Section 3

Item Number	Answer Key	Evidence Statement Key/ Content Scope
1.	B	3.NF.A.3a
2.	A	3.OA.C.7-2
3.	D	3.G.A.1
4.	C	3.NF.A.1
5.	A	3.OA.A.2
6.	B	3.MD.B.3
7.	647	3.NBT.A.2
8.	B	3.NF.A.3c
9.	B, D	3.OA.B.5
10.	C	3.MD.C.7d
11.	A	3.NF.A.2b

Section 4

Item Number	Answer Key	Evidence Statement Key/ Content Scope
1.	C	3.M.2 3.OA.A.3-2
2.	A, E	3.R.1 3.NF.A.3c
3.	<u>Sample Top Score Response</u> There are 9 supply boxes, and each box will need 6 colored markers, so the equation $6 \times 9 = 54$ means that the teacher needs a total of 54 colored markers to fill the supply boxes. The teacher needs 54 colored markers, and the teacher already had 15 colored markers, so the equation $54 - 15 = 39$ means that the teacher needs 39 more colored markers to fill the supply boxes. The teacher's thinking is correct. Refer to the Holistic Rubric for 3-Point Reasoning Constructed Response Items for score point information.	3.R.4 3.OA.A.3-1
4.	B	3.M.3 3.OA.A.1

Item Number	Answer Key	Evidence Statement Key/ Content Scope
5.	<u>Sample Top Score Response</u> The area of the smaller rectangle is $8 \times 6 = 48$ square feet. The area of the larger rectangle is $10 \times 9 = 90$ square feet. The area of both rectangles is $48 + 90 = 138$ square feet. Refer to the Holistic Rubric for 3-Point Modeling Constructed Response Items for score point information.	3.M.4 3.MD.D.8
6.	C, E	3.R.3 3.OA.B.5